Qatar-10b

R-0075

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-10_241119` · created 2026-07-15T19:44:42+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460633.649 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.043
Transit depth (Rp/Rs)^2	2.97 +/- 1.47 [%]
Orbital Inclination (inc)	82.02 +/- 2.58
Airmass coefficient 1 (a1)	1.069 +/- 0.014
Airmass coefficient 2 (a2)	-0.043 +/- 0.046
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	None
Optimal Aperture	2.93
Optimal Annulus	6.73
Transit Duration (day)	0.056 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.043 vs 0.1265 (deviation 0.91σ)
Duration fit vs published	0.056 d vs 0.1154 d (deviation 3.4σ)
Mid-time O–C	26.2 min
Depth SNR	3.4
Residual scatter	1.36 %

Light curve

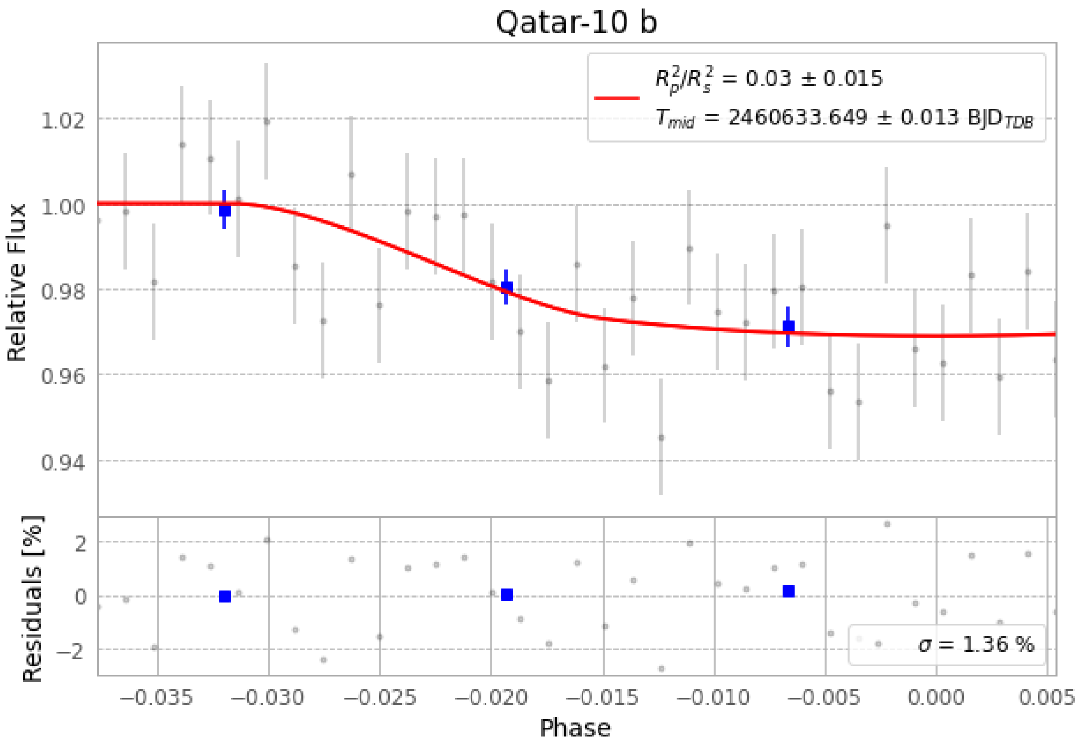

Figure 1 — FinalLightCurve_Qatar-10 b_2024-11-18.png (SHA-256 ce13731be13a8f61…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 295], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 599cc70c8cbd8fbf…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

35 FITS frames (23 MB), `Qatar-10241119020315.FITS` → `Qatar-10241119034516.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-241119.log (5a753c266427…), FinalLightCurve_Qatar-10 b_2024-11-18.png (ce13731be13a…), FinalLightCurve_Qatar-10 b_2024-11-18.csv (0782080ff444…)

Compute resources

Wall time 165.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-15 19:44Z.